



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JAYANTHI PETROLEUM AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

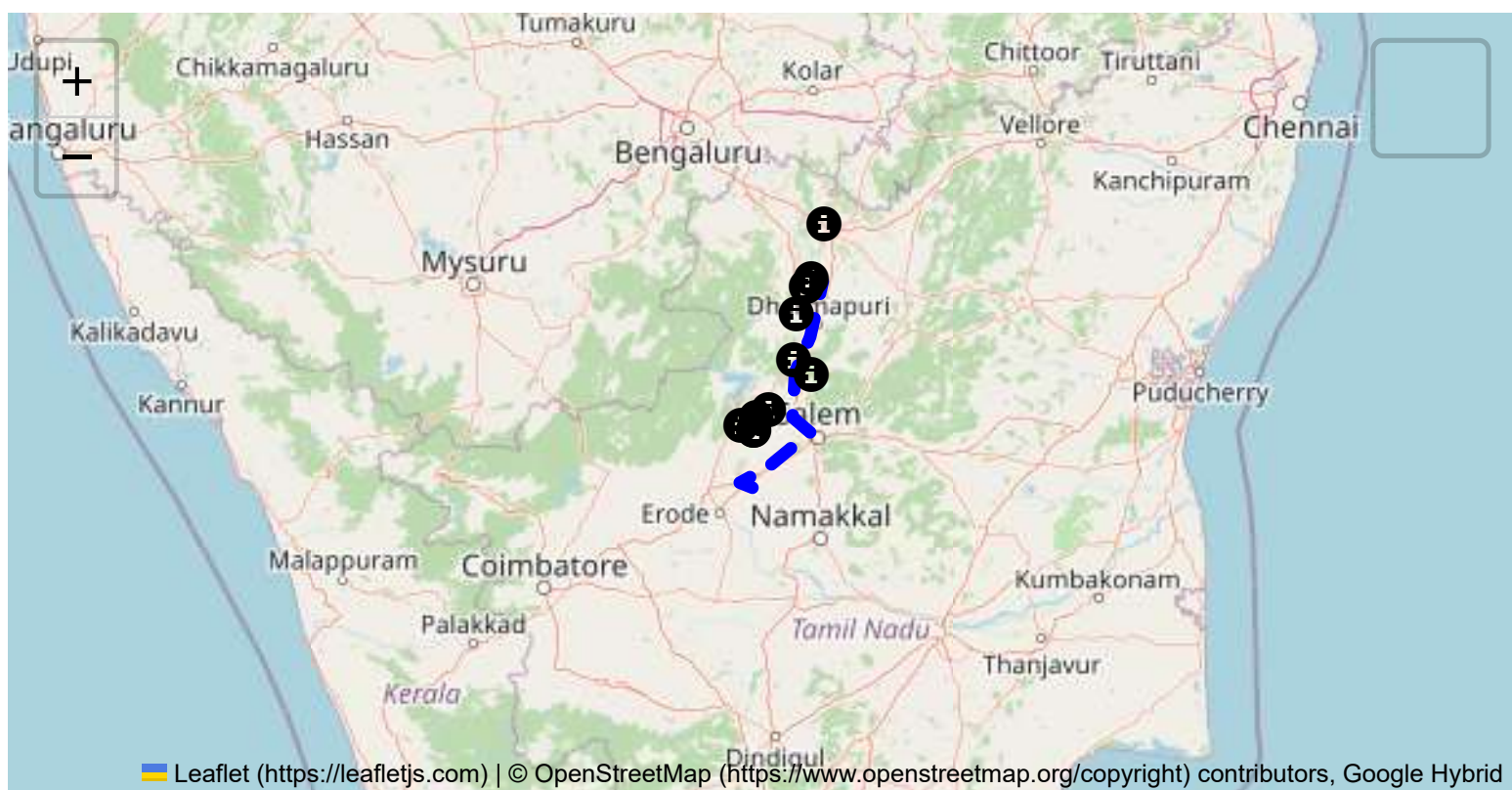
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

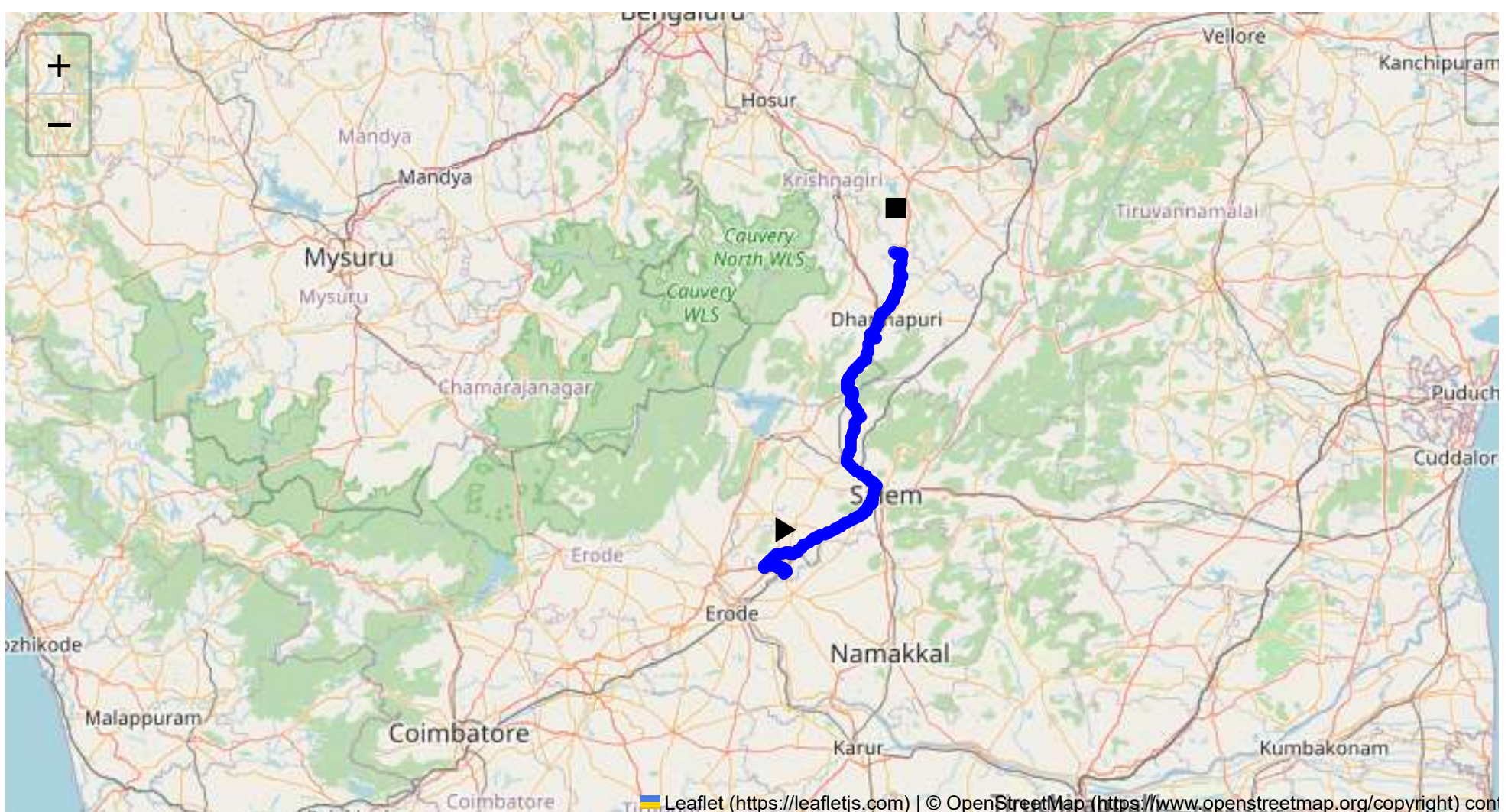
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 144.18 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.31729, 78.177567)



Welcome to the Journey Risk Management Study

Analyzing the described route from Sangagiri to Bommahalli in Tamil Nadu via several intermediate points provides a comprehensive insight into its safety and risks for truck drivers transporting hazardous materials. Here's an in-depth analysis considering the various factors of interest:

1. Overview of the Route Map

The route stretches approximately 144 kilometers and traverses various terrains and urban patches. Starting from Sangagiri, it includes a series of state and national highways, with transitions through both rural and urban regions, ending in Bommahalli. Key points are located along prominent roads, including Bhavani Main Road and the Salem-Kochi Highway.

2. Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. The Southwest monsoon (June to September) and Northeast monsoon (October to December) can significantly affect visibility and road conditions due to heavy rains, potentially causing flooding or landslides in some areas. This is a crucial consideration during the monsoon period when road surfaces may become slippery, increasing accident risks.

3. Traffic Patterns and Congestion

Peak congestion is typically observed during morning (7-9 AM) and evening (5-7 PM) hours around urban centers such as Salem. The bypass roads, although intended to alleviate urban traffic, can become bottlenecked with heavy vehicle traffic. Congestion is common near hospital areas, schools, and marketplaces along the route.

4. Road Quality and Infrastructure

The infrastructure quality varies along the route. In urban areas such as Salem, the road quality is generally good, with well-maintained surfaces. However, rural road segments may suffer from potholes and uneven surfaces, necessitating cautious driving. Signage and lighting may be inadequate in more remote areas.

5. Alternative Routes for Emergencies

In an emergency, diversions through local roads might be possible, particularly around Salem. It is advisable to have a GPS-enabled device for dynamic rerouting. However, detours through rural areas should be used cautiously due to potential road quality issues.

6. Local Regulations Affecting Hazardous Material Transport

Transporters must adhere to regulations on permissible timings for hazardous material transit, especially in populated areas. Restrictions on night travel (between specific hours), mandatory rest periods, and adherence to marked routes for such vehicles are enforced.

7. Historical Incidents

There have been incidents in the past involving heavy vehicles, often due to brake failure on steep descents and at junctions with inadequate signal systems. Local records may show occurrences of spills or leaks, emphasizing the need for rigorous safety checks.

8. Environmental Considerations

The route crosses regions of agricultural significance and potential water bodies. Care must be taken to prevent environmental contamination in the event of a spill, especially near sensitive areas or wildlife habitats.

9. Communication Coverage

Communication networks are generally robust in urban areas; however, coverage might be patchy in rural segments. Certain stretches may have signal dead zones, necessitating additional communication devices such as satellite phones.

10. Estimated Emergency Response Times

Response times vary along the route: in urban centers, emergency services can typically respond within 15-30 minutes, while in remote areas, it may take over an hour. It's crucial to note the locations of hospitals and fire stations en route.

12. Overall Summary of Risk Assessment

The route poses moderate risk levels, primarily due to weather-related hazards, variable road conditions, and traffic congestion. With adherence to regulations and strategic planning to avert peak traffic, risks can be managed effectively. Nonetheless, continuous monitoring of weather conditions and ensuring driver alertness, particularly during night drives and in low visibility conditions, remains paramount.

In conclusion, while transportation of hazardous materials along this route is feasible, it demands careful planning and adherence to safety protocols to mitigate identified risks effectively. Regular training for drivers on emergency response and ensuring vehicle readiness are critical for safe transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.73740, 78.05319	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44024, 77.87528	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45543, 77.81740	30 KM/Hr
8	Turn	High	11.45584, 77.81695	15 KM/Hr
9	Turn	Medium	11.45656, 77.81709	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.53552, 77.95763	30 KM/Hr
11	Turn	High	11.64158, 78.11796	15 KM/Hr
12	Turn	Medium	11.64310, 78.11982	30 KM/Hr
13	Turn	Medium	11.68022, 78.12748	30 KM/Hr
14	Turn	High	11.74252, 78.04705	15 KM/Hr
15	Turn	Medium	11.78027, 78.05791	30 KM/Hr
16	Turn	Medium	11.86967, 78.08387	30 KM/Hr
17	Turn	Medium	11.88030, 78.08150	30 KM/Hr
18	Turn	Medium	11.91289, 78.05854	30 KM/Hr
19	Turn	Medium	11.93237, 78.06459	30 KM/Hr
20	Turn	Medium	11.95594, 78.04730	30 KM/Hr
21	Turn	High	11.95876, 78.04990	15 KM/Hr
22	Turn	High	11.96035, 78.04946	15 KM/Hr
23	Turn	Medium	11.96304, 78.05136	30 KM/Hr
24	Turn	Medium	11.96399, 78.05135	30 KM/Hr
25	Turn	Medium	11.96458, 78.05084	30 KM/Hr
26	Turn	Medium	11.96496, 78.04988	30 KM/Hr
27	Turn	High	11.96580, 78.04923	15 KM/Hr
28	Turn	Medium	11.96675, 78.04939	30 KM/Hr
29	Turn	Medium	11.96791, 78.05070	30 KM/Hr
30	Turn	Medium	11.98490, 78.06275	30 KM/Hr
31	Blind Spot	Blind Spot	12.09202, 78.11459	10 KM/Hr
32	Turn	High	12.08780, 78.12237	15 KM/Hr
33	Turn	High	12.08688, 78.12234	15 KM/Hr
34	Blind Spot	Blind Spot	12.08465, 78.12762	10 KM/Hr
35	Turn	High	12.08688, 78.12234	15 KM/Hr
36	Turn	High	12.08780, 78.12237	15 KM/Hr
37	Turn	High	12.09202, 78.11459	15 KM/Hr
38	Turn	Medium	12.24590, 78.19512	30 KM/Hr
39	Turn	Medium	12.25185, 78.20031	30 KM/Hr
40	Turn	Medium	12.25477, 78.20000	30 KM/Hr
41	Turn	High	12.30916, 78.20317	15 KM/Hr
42	Turn	Medium	12.31000, 78.20220	30 KM/Hr
43	Turn	High	12.31077, 78.20193	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
13	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
16	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
20	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
21	hospital	Global Medical Centre Jagirammalayam	11.6894409, 78.1108847	30 km/h	Medium
22	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
24	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
25	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
27	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
28	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
29	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
30	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
31	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
32	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
33	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
34	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
35	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
36	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
37	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
38	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
39	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
40	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
41	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
42	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
43	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
44	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
46	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
48	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
49	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
50	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
52	pharmacy	Appolo	12.3088621, 78.2039069	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
14	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
19	school	Mothers Vidthyalaya Metric School	11.6884655, 78.1240569	30 km/h	Medium
23	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
26	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
45	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
47	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
51	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73740, 78.05319



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53552, 77.95763



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64158, 78.11796



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64310, 78.11982



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68022, 78.12748



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78027, 78.05791



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86967, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88030, 78.08150



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91289, 78.05854



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93237, 78.06459



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95594, 78.04730



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96035, 78.04946



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96304, 78.05136



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96675, 78.04939



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96791, 78.05070



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.98490, 78.06275



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.09202, 78.11459



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.08780, 78.12237



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.08688, 78.12234



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.08465, 78.12762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08688, 78.12234



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08780, 78.12237



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.09202, 78.11459



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24590, 78.19512



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25185, 78.20031



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25477, 78.20000



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.30916, 78.20317



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31000, 78.20220



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31077, 78.20193
